Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/19/2014
PI Name:	Karma, Alain Ph.D.		
Project Title:	Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	None		
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Comments:			
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COI Name (Institution):	Trivedi, Rohit (Iowa State University)		
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	Cellular and dendritic microstructures formed during industrial solidification processes (such as casting, welding, and brazing) determine the properties of commercial metallic alloys produced in numerous industries, with applications ranging from automobile engine parts to aircraft turbine blades. The production of innovative materials and the optimization of both properties and processing routes go through an accurate prediction and modeling of the dynamics of microstructures formation. From a fundamental standpoint, the formation of these structures is controlled by a subtle interplay between microscopic interfacial phenomena and macroscopic transport of heat and solute, which is strongly influenced by gravity-induced melt convection under terrestrial conditions. Therefore, microgravity directional Bridgman experiments provide a unique means to investigate microstructure development with negligible convection and to test the predictions of theoretical models in a purely diffusive growth regime.		

Task Description:	While most of the theoretical research to date has focused on the growth of single crystals, most commercially relevant metallic alloys are polycrystalline. The main aim of this research is to bring a quantum leap in our ability to model realistically dendritic alloy structures in polycrystalline materials where interactions between well-developed dendrites from the same grain and from grains of different crystal orientations both need to be considered. Interactions at grain boundaries are especially important because of their influence on dendrite fragmentation and the formation of solidification defects. This aim will be pursued by developing a new coarse grained dendritic network approach that is capable of simulating quantitatively the long-range diffusive interactions between primary, secondary and higher order branches of typical hierarchical dendritic tree structures on experimentally relevant length and time scales. Validation of this approach will be carried out through quantitative comparisons with phase-field simulations as well as both microgravity and ground based experiments in well-characterized aluminum based binary alloys, such as Al-Cu and Al-Si. Those studies will set the stage for planning future microgravity Bridgman experiments on the International Space Station that use bi-crystalline samples to investigate both intragrain and intergrain dendrite interactions in a diffusive growth regime. ED. NOTE (9/22/14): Project continues as "Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys--NNX14AB34G." See that project for subsequent reporting.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to light weight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. This research is aimed at developing computational tools to predict reliably the formation of these dendritic microstructures on experimentally relevant length and time scales so as to accelerate and optimize the design of new technological alloys that can directly benefit US industries.
Task Progress:	We have recently developed a novel dendritic needle network (DNN) model capable of simulating quantitatively the solidification of concentrated dendritic alloys with well developed primary, secondary, and higher order branches. This model makes it possible to simulate accurately the processing conditions used to solidify Al-7wt%Si alloy samples in the CETSOL microgravity experiments and Al-Cu alloys in the XRMON project. Our recent and current modeling efforts have mostly focused on developing a 3D extension of the DNN model. We have now extended the Dendritic Needle Network approach to 3D. The extension is not straightforward, since some of the basic assumptions of the 2D model rely on the analytical solution of the Laplace equation in the vicinity of a sharp needle tip, which has no equivalent solution in 3D. Hence, with the objective of formulating model similarly applicable to 3D, we first adapted the 2D DNN model to account for needle branches of finite thicknesses represented in the vicinity of the tip as parabola, in contrast to the initial assumption of sharp DNN needle-like branches. The calculation of the flux-intensity-factor at the tip – which captures the evolution of long range diffusion field seen locally at the tip of each dendrite – is now calculated using a Gauss contour integral around the tip, assuming a local Laplacian field in a frame attached to the dendrite tip, instead of the initial formulation using a contour integral similar to the Rice integral common in fracture mechanics. The resulting DNN implementation with parabolic needle branches of finite thickness is not only directly possible to extend to 3D, but since it modifies the solute field surrounding the tip, it also corrects the deviation of the steady-state growth from the analytical Ivantsov solution even when the supersaturation is not negligibly small, which was a limitation of the initial model formulation. The resulting DNN formulation with needles of finite thicknesses can readily be extended to three-dimensional modeling. The growth kinetics laws for needle tip radius rho and velocity V are modified in that the product rhoV now scales directly like the flux intensity factor F(t) calculated at the needle tip, thus replacing the law rhoV2 ~ F2 in 2D, while keeping the second assumption of rho2V ~ 1/sigma, where the tip selection parameter sigma is now given by 3D solvability theory. We achieved a first implementation of the DNN model in 3D, and validated the result against steady state and transient analytical solution of growth dynamics. For instance, for an idealized six-branches equiaxed grain at a given supersaturation, dimensional analysis imposes that the length of the branches L(t) grows with time t as ~ t2/3. We also validated that steady state growth matches the well know 3D Ivantsov solution. We are currently implementing the 3D DNN approach to non-isothermal conditions with multiple sidebranching events, to simulate the Al-7wt%Si directional solidification experiments achieved within the CETSOL project. The comparison with 2D DNN calculations will provide a good benchmark. Ultimately, the 3D DNN model should provide a powerful computationally efficient tool for dendritic solidification modeling, involving numerous grains and dendritic branches in 3D. The extension project NNX14AB34G builds on developments implemented in the framework of the present project, including an extension of the 2D DNN model to polycrystalline solidification to study the columnar-to-equiaxed transition, as well as making extensive use of the 2D DNN model to investigate mechanisms of complex diffusive interaction among dendritic branches, e.g. to propose new scaling laws for secondary dendritic arm coarsening. In parallel, we have implemented a quantitative phase-field (PF) model for directional solidification on massively parallel Graphic Processing Units (GPU) architecture, enabling the simulation of experimentally relevant length and time scale for dilute alloys. This approach complements the development and use of the multiscale DNN model, more adapted to simulate more concentrated alloys growing at a comparatively lower solute supersaturation. We have extended the 2D PF model to account for polycrystalline solidification, and used it extensively to study grain growth competition, through solutal interactions at converging and diverging grain boundaries upon columnar solidification. The resulting study highlights the stochasticity and complexity of grain boundary selection at the macroscopic scale during dendritic growth competition. Dendritic grain growth competition mechanisms are far more complex in three dimensions than in 2D, due to the additional degrees of freedom in a 3D configuration (e.g. relative position and orientation of the two competing grains). The computationally efficient GPU implementation of our phase-field simulation in 3D has already been achieved. In the framework of the project extension (award

	NNX14AB34G), we have extended the 3D PF model to bi-crystalline solidification, in a similar fashion as the 2D model. The validation of the 3D implementation with 3D orientation (i.e. with two Euler angles) is in progress and simulations of bi-crystalline columnar growth competition should follow. Our global efforts toward quantitative modeling of dendritic growth at a scale relevant to experiments will be carried on through a project extension – under award NNX14AB34G – through investigations on outstanding scientific topics such as the columnar-to-equiaxed transition, the dynamics of dendritic sidebranch coarsening at the macroscopic scale, and grain growth competition mechanisms in 3D. ED. NOTE (9/22/14): Project continues as "Integrated Computational and Experimental Studies of Complex Dendritic Microstructure Development during Directional Solidification of Metallic Alloys--NNX14AB34G." See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 05/10/2026)
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